

For Learning, Toil is Not Optional

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Recently, while browsing Reddit, I was served an ad by Anthropic—the creators of Claude—which claimed that "toil is optional." Ordinarily, I would ignore a line like that. But as a software developer, I recognise that it is not entirely hyperbole. As of late 2025, coding agents such as Claude Code demonstrate capabilities that would have been difficult to imagine even a few years ago.

Much of the routine burden of development—repeating familiar patterns, tracking down elusive bugs, maintaining documentation, and managing dependencies—is being reduced at a rapid pace.

In that sense, toil in software development is increasingly treated as a liability to be minimised. But in education, toil plays a different role. It is not incidental; it is formative. The effort involved in grappling with material is part of the biological processes through which understanding is built. When that struggle is offloaded, we alter the conditions under which mastery develops.

A 2025 study from the MIT Media Lab on AI-assisted writing found clear differences in brain activity:

EEG revealed significant differences in brain connectivity: Brain-only participants exhibited the strongest, most distributed networks; search engine users showed moderate engagement; and LLM users displayed the weakest connectivity.

Participants wrote an essay either unaided, with search engines, or with LLMs. While the study measures electrical activity, reduced engagement may correspond to reduced conditions associated with synaptogenesis and myelination—the

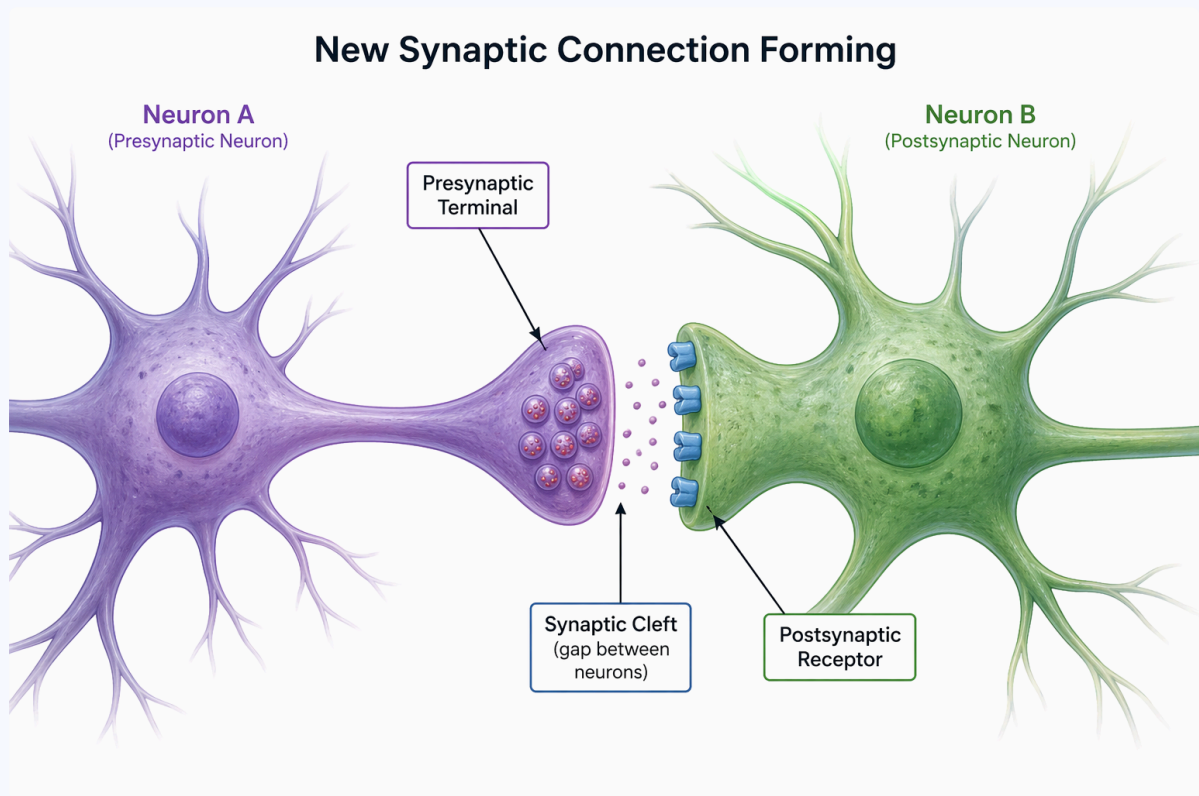
LLM (LARGE LANGUAGE MODEL)

A type of AI system trained on vast quantities of text to predict and generate language. LLMs learn statistical relationships between words, phrases, and concepts at scale. They do not "understand" in any cognitive sense — they model the distribution of language as it appears in training data. GPT-4, Claude, and Gemini are all LLMs.

processes by which neural connections are formed and strengthened through use.

Synaptogenesis

Synaptogenesis: the formation of new synaptic connections — the path begins to exist.

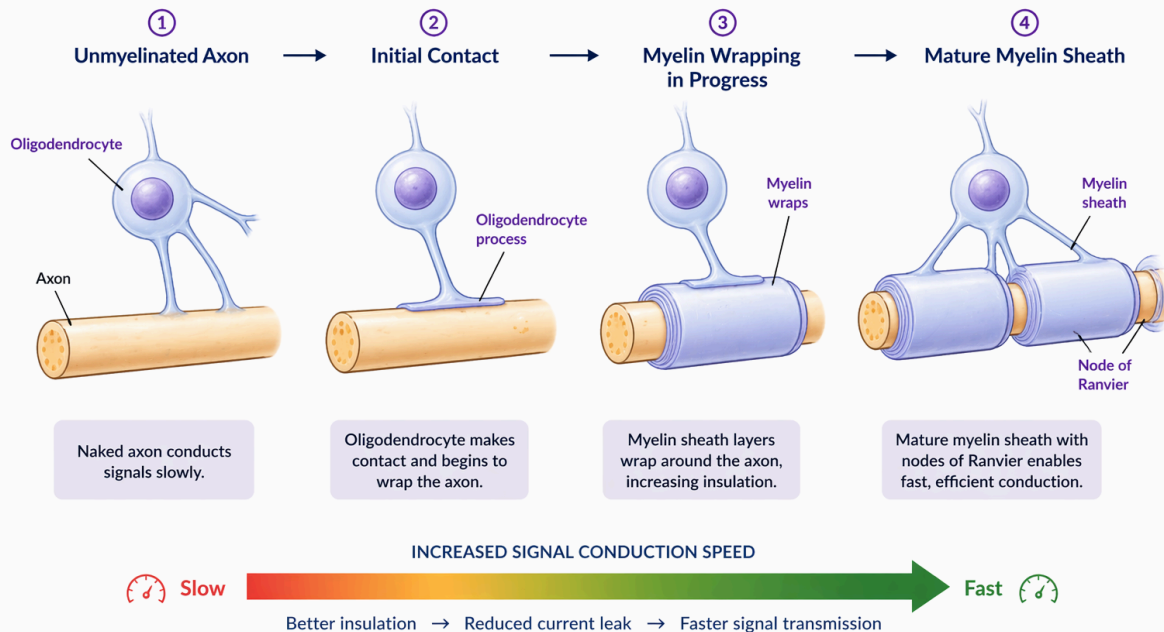


Myelination

Myelination: the insulation of neural pathways that makes transmission faster and more reliable — the path hardens.

Progressive Myelin Sheath Wrapping by Oligodendrocytes

Myelination increases signal conduction speed



Imagine a field of tall grass. Each time you walk between two landmarks—a boulder, a stream, a stand of trees—you press down a faint trail. That is synaptogenesis: the path begins to exist. Walk it repeatedly, and the trail hardens, the ground compacts, and movement becomes faster. That is myelination: the path becomes efficient.

A brain that labours lays down paths and reinforces them. A brain that offloads too early walks less ground—and builds fewer roads. Mastery requires walking those paths again and again. *Toil*, in other words.

Now consider LLMs. Imagine a machine traversing many possible paths between landmarks at a scale no human could approach. Modern training runs require a staggering 10^{26} discrete computations. This is a septillion-fold 'metabolic' effort—a relentless process of trial and error used to calculate and reinforce the digital pathways that eventually look like fluency.

Estimated discrete computations in training Grok-4

1 Septillion

$$10^{26} = 100,000,000,000,000,000,000,000,000$$

- Approximately the same as number of molecules in 3L of water
- Or 100 times the stars in the observable universe

This is the scale of repetition that underlies the model's apparent fluency. Through it, pathways between concepts are formed and reinforced at scale. Without that process, the model would have no reliable way to move between those landmarks at all.

So why do we think a student can skip theirs?

When we peek behind the curtain, the "magic" dissolves. What remains is scale—and toil. It feels effortless only because we never see the work that produced it.

That is the trap. Like a stage magician, the system presents the finished artifact while concealing the process that made it possible. If we hand a student that artifact—the essay—we are not accelerating their learning; we are bypassing it.

The paths are not walked. The trails are not cut. The roads are never consolidated. We may hold a finished piece of writing, but we remain in the field, without the means to move from one landmark to the next.

We cannot simply ban AI. The challenge is not exclusion, but design. We need to make **cognitive toil** desirable within environments where instant answers are always available.

This runs against the prevailing framing of such tools, where toil is treated as inefficiency to be eliminated. But the evidence suggests otherwise: when the struggle is removed, so too are the conditions under which learning is built.

The question is not whether difficulty can be avoided, but whether learning can survive without it. Human intelligence still appears to require a **metabolic price of admission**. Toil is that price.

COGNITIVE TOIL

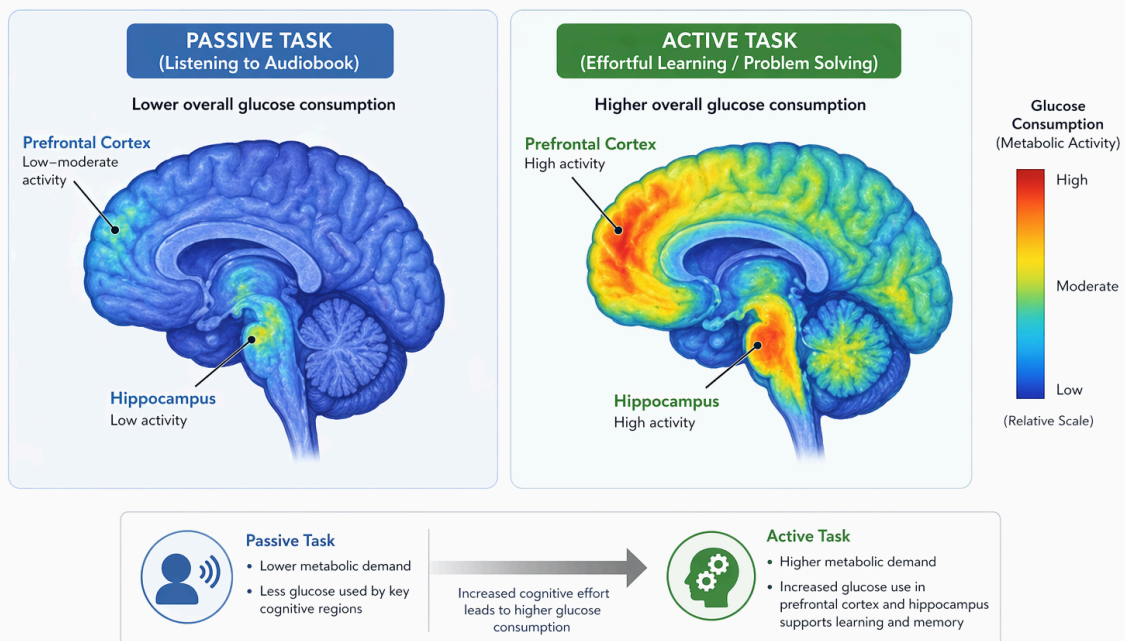
The effortful mental work involved in learning: retrieving information from memory, constructing explanations, resolving confusion, and producing original responses to novel problems. Cognitive toil is the mechanism through which knowledge becomes durable and transferable — not a sign that instruction has failed, but evidence that it is working. The term contrasts with task completion, which can occur without any of this effort when AI generates the output on the learner's behalf.

metabolic price of admission

Learning has a literal metabolic cost — glucose consumption rises during effortful cognition, and that cost cannot be skipped.

Brain Glucose Consumption: Passive vs. Active Cognitive Tasks

Effortful learning increases metabolic activity, especially in prefrontal and hippocampal regions.



Images represent relative glucose metabolism measured by FDG-PET.